

GENETICS

A temperature-tolerant CRISPR base editor mediates highly efficient and precise gene editing in *Drosophila*

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CRISPR nucleases generate a broad spectrum of mutations that includes undesired editing outcomes. Here, we develop optimized C-to-T base editing systems for the generation of precise loss- or gain-of-function alleles in *Drosophila* and identify temperature as a crucial parameter for efficiency. We find that a variant of the widely used APOBEC1 deaminase has attenuated activity at 18° to 29°C and shows considerable dose-dependent toxicity. In contrast, the temperature-tolerant evoCDA1 domain mediates editing of typically more than 90% of alleles and is substantially better tolerated. Furthermore, formation of undesired mutations is exceptionally rare in *Drosophila* compared to other species. The predictable editing outcome, high efficiency, and product purity enables near homogeneous induction of STOP codons or alleles encoding protein variants in vivo. Last, we demonstrate how optimized expression enables conditional base editing in marked cell populations. This work substantially facilitates creation of precise alleles in *Drosophila* and provides key design parameters for developing efficient base editing systems in other ectothermic species.

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INTRODUCTION

The manipulation of genes through the induction of precise mutations is a central tool to establish causal genotype-phenotype relationships. Over the last decade, clustered regularly interspaced short palindromic repeats–Cas (CRISPR-Cas) systems have emerged as the preferred system for targeted genome manipulation, mainly due to their high efficiency and ease of use. CRISPR nucleases, such as Cas9 or Cas12a, can be programmed by guide RNAs (gRNAs) to create DNA double-strand breaks (DSBs) at defined genomic loci, which are resolved by error-prone endogenous DNA repair pathways, thereby creating mutations at the target site (1–3). A pervasive shortcoming of this strategy is the uncontrollable nature of the induced DNA alterations, which are typically dominated by short insertions and deletions (indels) (4), but also include large indels and genomic rearrangements at considerable frequency (5, 6). Typically, not all of the induced mutations have the desired effect. For example, small in-frame indels can subvert gene disruption efforts as they often do not interfere with the function of the encoded protein (4, 7, 8), and larger deletions and rearrangements can cause false-positive phenotypes by affecting genes other than the target. Therefore, methods that can efficiently generate precise mutations would offer advantages for functional genomics.

CRISPR base editors are fusion proteins of a Cas9 nickase linked to a deaminase domain and can be used to install defined point mutations in the genome (9–11). Cytosine base editors (CBEs) create C-to-T edits through (i) R-loop formation, (ii) deamination of cytosine residues in the single-stranded, nontarget DNA strand by a cytidine deaminase to convert cytosine to uracil, and (iii) propagation of the edit through DNA repair and/or cell division (fig. S1, A

and B) (12). DNA deamination occurs in a specific interval within the R-loop, called the editing window. As a result, cytosines other than the target residue are likely to be edited if they are present within the editing window (12). While these so-called bystander edits are often problematic when the aim is to install or correct missense variants, a sizable fraction of biologically or clinically relevant missense variants can be introduced through base editing without nonsilent bystander edits (13). Moreover, CBE-based gene inactivation strategies, such as converting codons encoding amino acids into stop codons (14, 15) or mutating splice sites (16, 17), are generally not limited by the occurrence of bystander edits.

Predictable mutations introduced by base editing could circumvent some of the limitations associated with random mutations introduced by CRISPR nucleases, provided that they are installed with high efficiency and minimal undesired by-products. However, several shortcomings and uncertainties exist. First, while CBEs and other base editors have been extensively optimized for expression, nuclear localization (18, 19) and activity of the deaminase domain (12, 20), experiments aiming to obtain homogeneous somatic phenotypes require particularly high editing rates, which remain difficult to attain with base editors in various systems (21–23). This may at least partially be due to the fact that genome editing tools are typically developed and optimized in immortalized mammalian cell lines, which differ from many other systems relevant for research and biotechnology in parameters such as temperature and composition of DNA repair pathways. Second, while CBEs predominantly give rise to predictable C-to-T edits in the editing window, they are also known to cause random indel mutations through error-prone repair of abasic sites that arise from uracil removal by uracil-*N*-glycosylases (UNG) (10, 12), a process which can only partially be suppressed through UNG inhibitor (UGI) domains (20, 24, 25). Induction of such random indels limits the ability of CBEs to avoid undesirable editing products.

Drosophila melanogaster is one of the most widely used model organisms in biomedical research, largely due to the availability of sophisticated genetic tools (26). Over the years, genetic research in *Drosophila* has uncovered numerous fundamental and evolutionary

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conserved principles governing the biology of multicellular animals (27–29). Despite these efforts, functionally validated null alleles are still not available for the majority of genes encoded in the *Drosophila* genome (26). In recent years, large collections of CRISPR gRNA lines have been generated that can be used in conjunction with Cas9 nuclease to induce targeted mutations (8, 30–32), but these suffer from the aforementioned limitations of random mutagenesis. So far, only a single study has reported on the use of a first-generation base editor in *Drosophila*, which was shown to induce C-to-T edits in less than half of the tested genomic target sites (22). Whether optimized base editors can lead to more robust editing in this organism remains to be explored.

Here, we establish CBEs for gene editing in *Drosophila* and uncover crucial parameters influencing the performance of these tools. While the efficiencies obtained with a derivative of the most commonly used apolipoprotein B mRNA editing enzyme catalytic subunit 1 (APOBEC1) domain were constrained by the temperature range tolerated by *Drosophila*, a domain derived from an ectothermic species living in cold to temperate environments permitted robust base editing with remarkably high efficiency. Moreover, we uncovered unusually high CBE product purity in *Drosophila*, which is likely attributable to the lack of a *UNG* gene in holometabola (33). We show that our tools permit highly homogeneous loss-of-function mutagenesis and, unlike nucleases, can furthermore efficiently generate defined missense variants in both soma and germline. Our results highlight the usefulness of CBEs for gene editing in multicellular organisms, establish robust and highly efficient tools for *Drosophila*, and reveal so far underappreciated parameters that influence the activity and purity of base editing systems.

RESULTS

Differential temperature sensitivity of the *evoAPOBEC1* and *evoCDA1* cytidine deaminases

To explore CBE-mediated gene editing in *Drosophila*, we first constructed vectors for in vivo CBE expression. We used a Cas9^{D10A} coding sequence previously shown to be well tolerated in *Drosophila* (34), linkers and bipartite nuclear localization signals from the optimized BE4max architecture (18), as well as a UGI domain (Fig. 1A) (10). To explore the properties of two distinct deaminases, we constructed CBEs featuring recently laboratory evolved versions of the APOBEC1 protein from *Rattus norvegicus* and the cytidine deaminase 1 (CDA1) protein from *Petromyzon marinus*, respectively (20) (henceforth referred to as *Drosophila* CBE^{evoAPOBEC1/evoCDA1}, dCBE^{evoAPOBEC1}, and dCBE^{evoCDA1}).

For ubiquitous base editing, we cloned these editors downstream of the *act5c* promoter (*pAct-dCBE^{evoAPOBEC1}* and *pAct-dCBE^{evoCDA1}*; fig. S2A) and generated transgenic animals harboring genomic insertions of these constructs on a defined landing site on the second chromosome. Such animals were viable, fertile, and without any obvious phenotypic abnormalities and can be maintained as a stable stock under standard conditions (fig. S2B).

Next, we set out to assess the efficiency of CBE-mediated gene inactivation. To facilitate both phenotypic and genotypic readouts, we initially targeted a set of nonessential genes with well-characterized, visible, and recessive null mutant phenotypes [*ebony* (*e*), *singed* (*sn*), *forked* (*f*), and *sepia* (*se*)]. To this end, we used the previously described *pCFD5* vector to ubiquitously express pairs of

gRNAs (35) designed to either mediate the conversion of codons coding for amino acids to STOP codons (14, 15) or to disrupt splice donor or acceptor sites (17) at two independent positions in each gene (Fig. 1B). To establish optimal conditions for base editing in *Drosophila*, we crossed males from four gRNA strains (one targeting *se*, two targeting *e*, and one targeting *sn*) against *pAct-dCBE^{evoAPOBEC1}* or *pAct-dCBE^{evoCDA1}* virgins ubiquitously expressing either CBE, raised the animals at various temperatures within the range used to maintain *Drosophila* (18°, 24°, and 29°C; Fig. 1B) and assessed offspring phenotypically and by deep sequencing of PCR amplicons (Amp-Seq) of the target sites.

We observed strong loss-of-function phenotypes when base editing was performed with dCBE^{evoAPOBEC1} at 29°C, with fully penetrant phenotypes observed with three gRNA strains and a mosaic phenotype with one strain (Fig. 1C). However, phenotypes were attenuated when animals were raised at 24°C and nearly absent at 18°C (Fig. 1C). In contrast, dCBE^{evoCDA1} mediated fully penetrant phenotypes resembling heritable, homozygous loss-of-function alleles in all instances at 29° and 24°C. Unexpectedly, even at 18°C, animals developed phenotypes indicative of biallelic gene disruption in most or all of their body.

We then used amplicon sequencing to sensitively quantify mutations at the target site of each gRNA. Consistent with our phenotypic observations, we found a strong correlation between temperature and C-to-T editing rates with dCBE^{evoAPOBEC1}, but highly efficient editing at at least one target site across all conditions with dCBE^{evoCDA1} (Fig. 1C). At the target residue, C-to-T editing rates at 29°C were $51.9 \pm 30\%$ with dCBE^{evoAPOBEC1} compared to $89.8 \pm 16.5\%$ with dCBE^{evoCDA1}, $29.7 \pm 20.3\%$ compared to $82.4 \pm 26.5\%$ at 24°C, and $10.4 \pm 8.8\%$ compared to $73.6 \pm 31.8\%$ at 18°C. Summarizing across all gRNAs and all C nucleotides within an editing window of positions 3 to 9 of the protospacer, base editing efficiency decreased an average of 5.7-fold from 29° to 18°C with dCBE^{evoAPOBEC1} but only 1.3-fold with dCBE^{evoCDA1} (Fig. 1D), demonstrating that activity of the *evoCDA1* deaminase domain is much less temperature sensitive at that range.

Together, these findings demonstrate that the *evoAPOBEC1* domain functions suboptimally at the lower temperatures inhabited by *Drosophila* and other ectothermic organisms, whereas *evoCDA1* permits efficient base editing within this broad temperature range.

dCBE^{evoCDA1} permits highly efficient biallelic gene inactivation in vivo

To further explore whether dCBE^{evoCDA1} can mediate gene knock-out with high penetrance in somatic cells at a larger selection of target sites, we generated six additional pairs of gRNAs targeting the aforementioned nonessential genes, as well as three pairs of gRNAs designed to create STOP codons in well-characterized essential genes [*cut* (*ct*), *evenness interrupted* (*evi*), and *smoothened* (*smo*)].

In crosses targeting the nonessential genes at 29°C, *pAct-dCBE^{evoCDA1}* led to fully penetrant loss-of-function phenotypes in all but one instance (Fig. 2A and fig. S3A). In the case of *f^{splice_a}*, where gRNAs target the splice acceptors of two small exons, we did not observe any phenotype indicating disruption of *f*. This could reflect inefficient base editing or result from biological compensation by exon skipping (fig. S4A) (16). Excluding the former possibility, we observed highly efficient ($90.0 \pm 2.25\%$ and $90.0 \pm 1.36\%$) C-to-T conversion at the target residue of both gRNAs present in

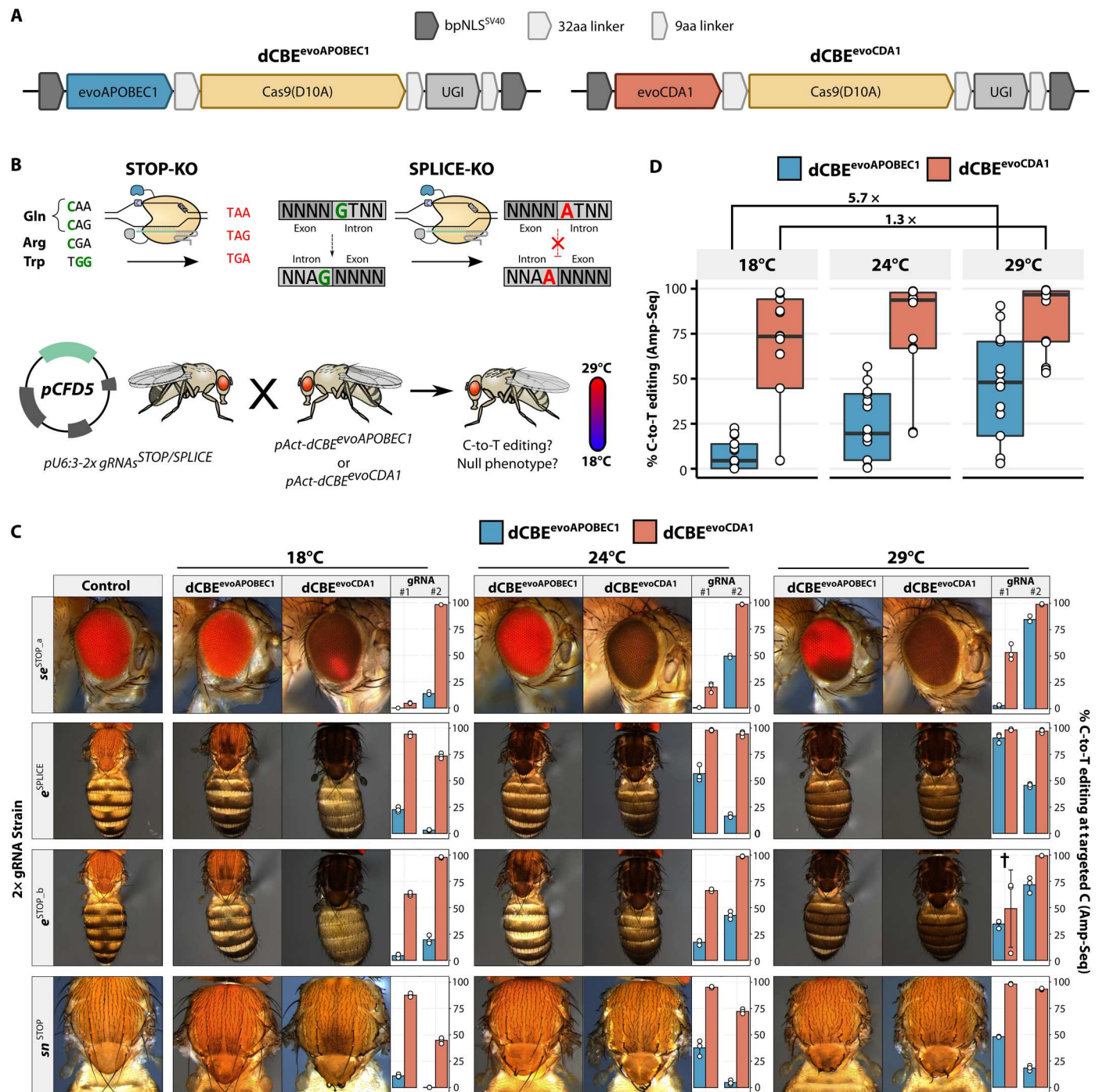


Fig. 1. CBE-mediated gene inactivation across the thermal range tolerated by *Drosophila*. (A) Schematic representation of the base editor constructs used in this study. (B) Schematic representation of the experimental workflow. Pairs of gRNAs were designed to cause C-to-T edits either by disrupting splice sites [SPLICE-knock out (KO)] or generating de novo STOP codons [STOP-knock out (KO)]. Males ubiquitously expressing the gRNAs were crossed to base editor-expressing virgins, and the offspring raised at different temperatures. (C) Representative images of female offspring are shown alongside the C-to-T editing rates at the targeted C residues, determined by Amp-Seq ($n = 3$, data presented as mean $\pm$ SD (bars with error bars) and individual measurements (points)). Whereas editing and phenotypic severity are strongly correlated with temperature for *pAct-dCBE^{evoAPOBEC1}*, *pAct-dCBE^{evoCDA1}* mediates efficient gene inactivation across all conditions. The target site for the gRNA marked with a cross was found to harbor a floating polymorphism located two nucleotides upstream of the PAM, which was homozygous in one fly analyzed for *dCBE^{evoCDA1}* at 29°C, giving rise to the outlying data point (see also fig. S4B). gRNAs are named according to their target gene, whether they mediate the formation of stop codons or disrupt splice sites and a small letter in case there are more than one gRNA for each class. (D) Effect of temperature on base editing efficiency. Each dot represents the mean (from $n = 3$) of a C residue located within positions 3–9 of any of the assessed target sites (8 gRNAs, 20 C residues). Horizontal line represents mean, hinges the first and third quartiles, and whiskers 1.5 interquartile ranges. Fold changes in average base editing efficiency between 18° and 29°C are indicated. UGI, uracil DNA glycosylase inhibitor.

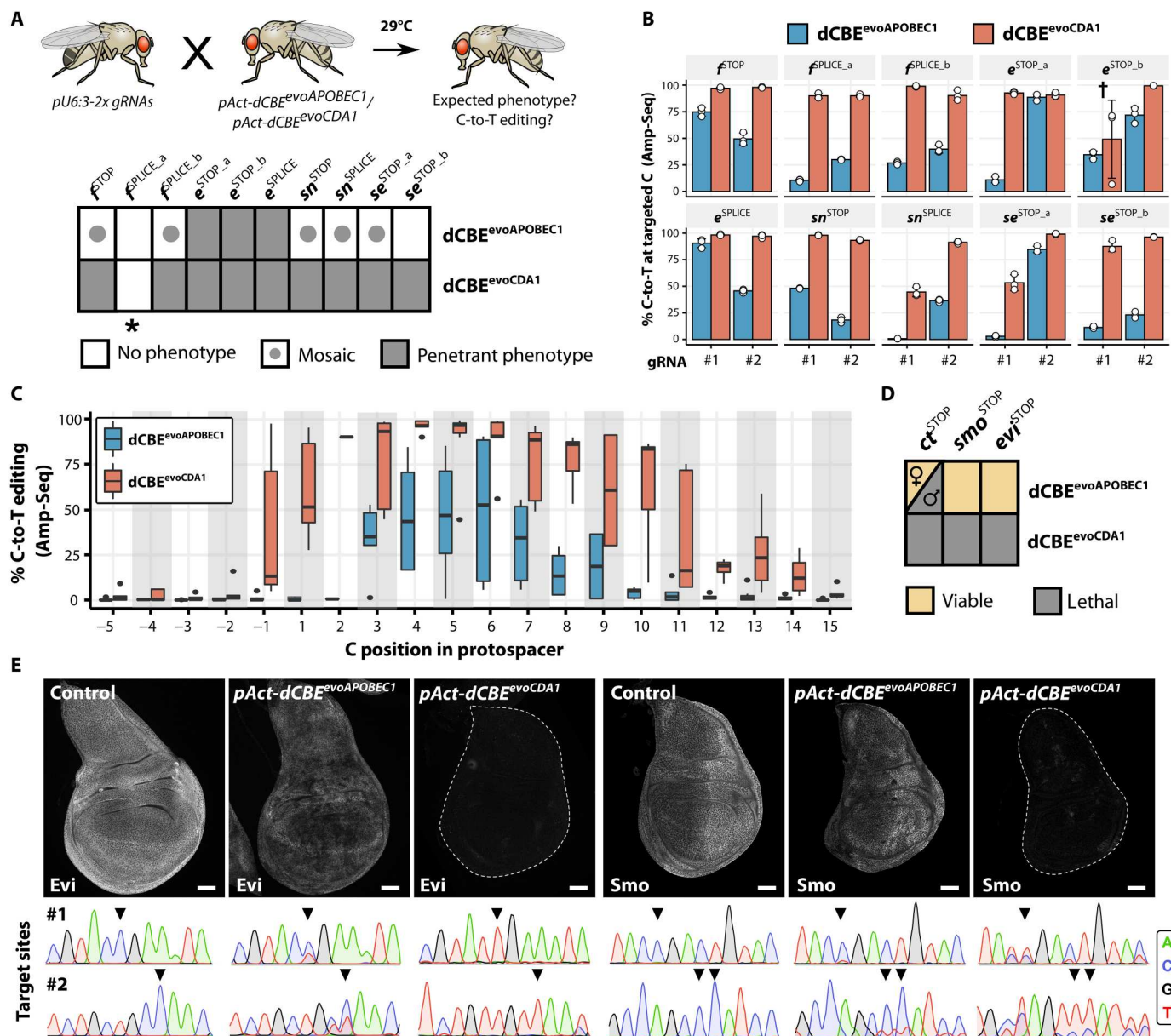


Fig. 2. Highly efficient biallelic gene inactivation with dCBE^{evoCDA1}. (A) Schematic representation of phenotypes at 29°C. While outcomes with *pAct-dCBE^{evoAPOBEC1}* are heterogeneous, *pAct-dCBE^{evoCDA1}* causes penetrant phenotypes with all gRNA pairs except *f^{SPICE_a}* (asterisk), likely caused by exon skipping (see fig. S3A for images of phenotypes). (B) C-to-T editing rates at the intended C residues, determined by Amp-Seq [$n = 3$, data presented as mean $\pm$ SD (bars with error bars) and individual measurements (points)]. *dCBE^{evoCDA1}* achieves highly efficient C-to-T editing at one or both gRNAs for all pairs, including *f^{SPICE_a}*. Target site of the gRNA marked with a cross harbors a floating polymorphism, which was homozygous in one fly analyzed for *dCBE^{evoCDA1}* at 29°C, likely reducing efficiency. (C) Editing efficiency of *pAct-dCBE^{evoAPOBEC1}* and *pAct-dCBE^{evoCDA1}* at 29°C plotted as a function of position in the protospacer ($N = 20$ target sites, individual data points are the mean from $n = 3$ Amp-Seq measurements for each target site). Horizontal line represents mean, hinges first and third quartiles, and whiskers 1.5 interquartile ranges. *dCBE^{evoCDA1}* has superior base editing efficiency and a broader editing window compared to *dCBE^{evoAPOBEC1}*. (D) Mutagenesis of the essential genes *ct*, *evi*, and *smo*. gRNA males were crossed to *pAct-dCBE^{evoAPOBEC1}* or *pAct-dCBE^{evoCDA1}* virgins and raised at 29°C. While offspring of *dCBE^{evoAPOBEC1}* crosses are mostly viable, *dCBE^{evoCDA1}* produces the expected outcome of lethality. (E) Immunofluorescence images and Sanger sequencing of third instar larvae subjected to ubiquitous mutagenesis with *evi^{STOP}* and *smo^{STOP}* gRNAs. While *pAct-dCBE^{evoAPOBEC1}* results in mosaic wing imaginal discs, *pAct-dCBE^{evoCDA1}* leads to penetrant protein ablation. Sanger chromatograms exhibit incomplete C-to-T editing with *pAct-dCBE^{evoAPOBEC1}*, but efficient conversion with *pAct-dCBE^{evoCDA1}*. Scale bars, 50 μ m. Triangles highlight the target C residue. Note that there is also detectable base editing at other C nucleotides (see also fig. S6).

SPLICE_{a} (Fig. 2B). Together, these results demonstrate that $\text{dCBE}^{\text{evoCDA1}}$ not only mediates phenotypes with remarkably high penetrance and robustness, but also highlights that the outcome of splice acceptor disruption is not always predictable (e.g. can lead to both intron retention or exon skipping), as has also previously been observed (16, 17).

In contrast to our results with $\text{dCBE}^{\text{evoCDA1}}$, phenotypic outcomes resulting from editing with $\text{dCBE}^{\text{evoAPOBEC1}}$ were much more heterogeneous. We observed fully penetrant phenotypes in three instances, mosaic phenotypes with five gRNA pairs, and no detectable phenotype in two conditions (Fig. 2A and fig. S3B). Sequencing of the gRNA target sites confirmed highly efficient C-to-T conversion of the target deoxycytidine with $\text{dCBE}^{\text{evoCDA1}}$, but large variation in base editing efficiency at the target base as the underlying cause of variation in phenotypic penetrance with $\text{dCBE}^{\text{evoAPOBEC1}}$ (Fig. 2B). We tested whether a publicly available machine-learning model could predict which gRNAs support efficient editing with $\text{dCBE}^{\text{evoAPOBEC1}}$, but found only a modest and not statistically significant correlation between the Z-score predicted by the model (36) and the observed fraction of base-edited alleles at each target site (Pearson $R = 0.34$; fig. S5).

To determine the average base editing efficiency across the base editing window at 29°C with both editors, we calculated the averaged C-to-T conversion rates as a function of position in the protospacer. In line with previous reports for these deaminases (20), $\text{dCBE}^{\text{evoAPOBEC1}}$ and $\text{dCBE}^{\text{evoCDA1}}$ exhibited editing windows of different size, spanning positions 3 to 9 and -1 to 14, respectively (Fig. 2C, defined as >10% median C-to-T editing efficiency). While editing rates were generally highest at the target C, most sites showed substantial editing at bystander C nucleotides, in particular with $\text{dCBE}^{\text{evoCDA1}}$ (fig. S6). Overall, $\text{dCBE}^{\text{evoAPOBEC1}}$ achieved a maximal mean editing efficiency of 45.9 to 53.5% at positions 4 to 6 of the protospacer, while $\text{dCBE}^{\text{evoCDA1}}$ enabled highly robust C-to-T conversion of 88.3 to 96.3% at the same positions of the target sites.

Next, we crossed the *pAct-dCBE* constructs to gRNA pairs targeting the essential genes *smo*, *evi*, and *ct*. While $\text{dCBE}^{\text{evoCDA1}}$ /gRNA animals died during development, the known loss-of-function phenotype of these target genes, crosses with $\text{dCBE}^{\text{evoAPOBEC1}}$, gave rise to viable offspring with wing malformations (Fig. 2D and fig. S7), indicating inefficient mutagenesis. To confirm this, we stained wing imaginal discs from third instar larvae from crosses targeting *evi* and *smo* for endogenous Evi and Smo protein and assayed the status of each target site by Sanger sequencing. While $\text{dCBE}^{\text{evoAPOBEC1}}$ caused incomplete loss of protein expression in the wing disc and led to limited editing of the target residues, $\text{dCBE}^{\text{evoCDA1}}$ mediated almost complete ablation of Evi and Smo in all cells of the tissue and caused efficient C-to-T editing at the targeted C residues (Fig. 2E and fig. S6). Together, these results establish that the *evoCDA1* domain permits highly efficient and robust C-to-T editing in *Drosophila*, resulting in penetrant phenotypes that allow for facile phenotyping in F1 animals.

Base editing by-product formation is exceptionally rare in *Drosophila*

A key advantage of base editing over CRISPR nuclease-mediated genome editing lies in its predictable editing outcomes. However, recent studies in mammalian cells and other systems have highlighted the occurrence of undesired mutations, including indels and C-

to-G or C-to-A edits, induced by CBEs (10, 20, 24). The frequency and occurrence of such by-products in *Drosophila* remain unknown. To address this, we examined our highly sensitive Amp-Seq dataset to determine the presence of indel by-products in flies subjected to mutagenesis using *pAct-dCBE^{evoAPOBEC1}* or *pAct-dCBE^{evoCDA1}* at 29°C.

The average indel frequencies with $\text{dCBE}^{\text{evoAPOBEC1}}$ and $\text{dCBE}^{\text{evoCDA1}}$ were 0.5 and 2.3%, respectively, which is substantially below the indel rates reported using the same deaminases in human cells (8.9 and 12.4%, respectively; Fig. 3A) (20). Likewise, C-to-G or C-to-A edits were virtually absent at the vast majority of target sites (Fig. 3B), despite the very high C-to-T editing rates (see Fig. 2B). The indels we did observe were highly diverse and frequently comprised the editing window (fig. S8A), suggesting deaminase rather than Cas9 nickase activity as causative for these edits. Supporting this hypothesis, target sites with four C residues in their editing window tended to show higher indel formation rates than sites with fewer C nucleotides (fig. S8B).

To control for potential differences in the overall base editing activity between our experiments and other studies, we calculated the C-to-T:indel ratio for C residues with at least 5% base conversion at any given target site. With both $\text{dCBE}^{\text{evoAPOBEC1}}$ and $\text{dCBE}^{\text{evoCDA1}}$ most C residues exhibited C-to-T:indel ratios between 30 and 300 (Fig. 3C). In contrast, other studies using *evoAPOBEC1* and *evoCDA1* in immortalized human cells (20), mouse embryonic fibroblasts (37), medaka, or zebrafish (38) reported C-to-T:indel ratios between 5 and 25 for *evoAPOBEC1* and between 2 and 16 for *evoCDA1* (Fig. 4C). This reveals that base editing-induced indel formation is exceptionally rare in *Drosophila*. Intriguingly, *Drosophila* and other Holometabola lack a *UNG* gene (33), suggesting a mechanism where Uracil tolerance leads to reduced indel formation and increased installation of desired C-to-T edits in these species (Fig. 4D), which is in line with a previous study showing substantially improved CBE product purity in human *UNG* knockout cells (24). The few observed indels might be caused by other components of base excision repair (BER), many of which lead to by-products in human cells (39) and have orthologs in *Drosophila* (40). In summary, these analyses show that aside from achieving high C-to-T conversion rates, CBEs in *Drosophila* also exhibit remarkably elevated product purity and suggest that current strategies for *UNG* inhibition in other species are only partially successful.

Creating heritable alleles with base editing in the *Drosophila* germ line

While a previous study established base editing with a first-generation CBE in *Drosophila* somatic cells (22), germline transmission of base edits has thus far not been described. We first performed a genetic complementation assay with both of our editors by crossing *pAct-dCBE^{evoAPOBEC1}*; *pCFD5-2xgRNA* or *pAct-dCBE^{evoCDA1}*; *pCFD5-2xgRNA* males raised at 29°C to virgins harboring homozygous null alleles in the targeted genes and assessed the percentage of offspring inheriting two loss-of-function alleles and thus exhibiting the respective null mutant phenotype (Fig. 3A). For all four assayed gRNA lines, we detected germline transmission rates that were in line with the somatic editing efficiencies we had observed previously (75, 70, 95, and 84% for *pAct-dCBE^{evoAPOBEC1}* and 100, 96, 97, and 96% for *pAct-dCBE^{evoCDA1}*, respectively; Fig. 3B).

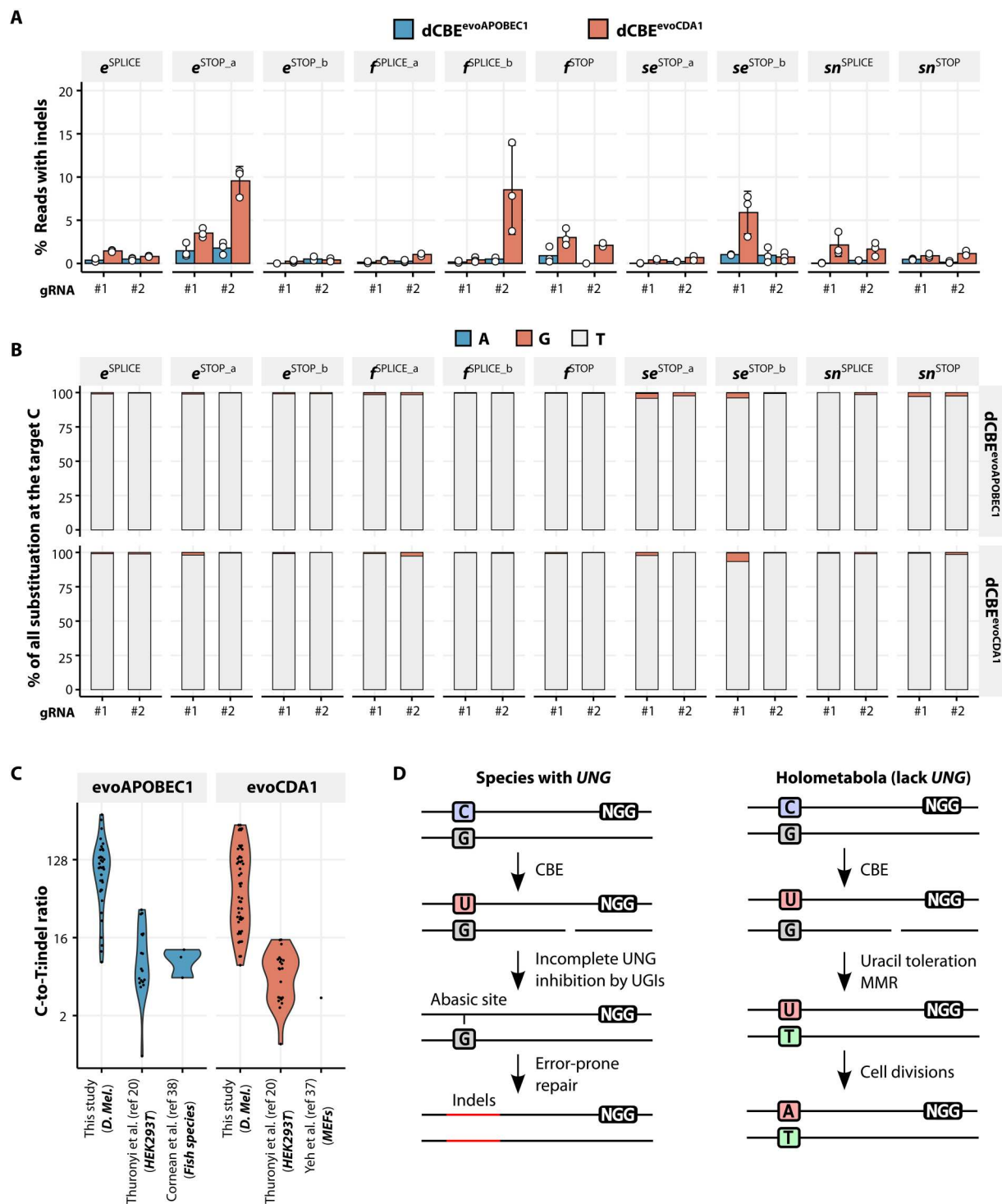


Fig. 3. Analysis of by-products caused by cytosine base editing. (A) Percentage of Amp-Seq reads containing indels at 29°C ($n = 3$, data presented as mean $\pm$ SD and individual measurements). With some exceptions, indel formation is generally low ($<2\%$ at most target sites) and slightly higher with dCBE^{evoCDA1} compared to dCBE^{evoAPOBEC1}. (B) Frequency of different types of base substitution observed at the target C residues. With both deaminases and across the vast majority of target sites, virtually all observed substitutions are C-to-T, with only a few target sites exhibiting low levels of C-to-G conversion (data presented as the mean frequency of reads exhibiting the respective substitution out of all reads harboring a non-C nucleotide at the target residue). (C) C-to-T over indel ratio observed with evoAPOBEC1 and evoCDA1 in this and other studies. Shown are the ratios for all C-residues at positions 3 to 10 of the target sites that exhibit at least 5% C-to-T conversion within the respective datasets. Compared to observations in other systems, the C-to-T:indel ratios are drastically elevated in *Drosophila*. (D) Proposed mechanism for reduced indel formation in *Drosophila*. In species harboring a *UNG* gene, incomplete inhibition of the enzyme by UGI domains results in formation of abasic sites, which are repaired in an error-prone fashion. In *Drosophila* and presumably other holometabola, Uracil is tolerated in genomic DNA (gDNA), favoring mismatch repair and resolution to a desired C-to-T edit. MEFs, mouse embryonic fibroblasts.

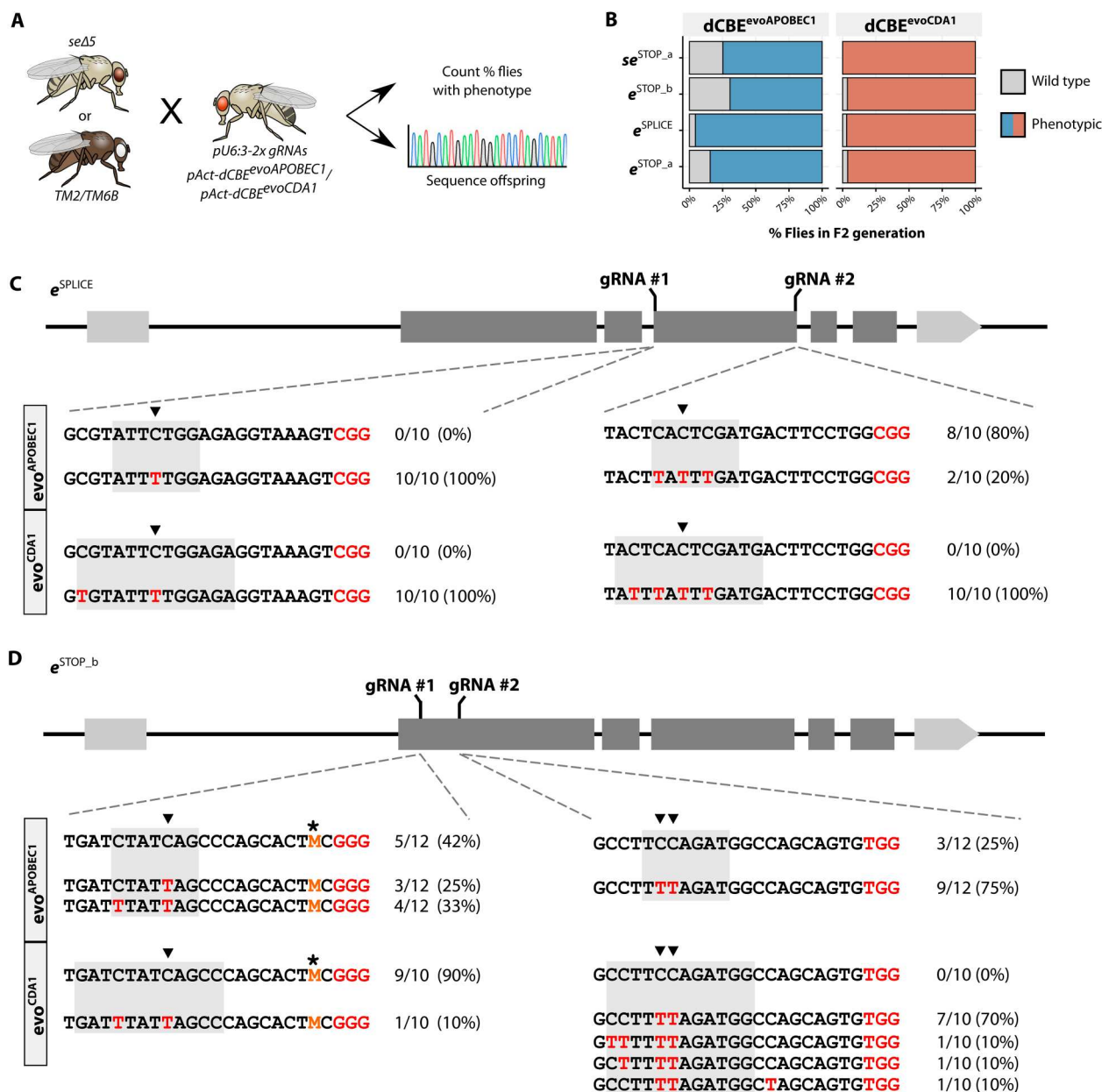


Fig. 4. Germline transmission of base-edited alleles. (A) Schematic of the genetic complementation assay for germline transmission. F1 males harboring both genome editing transgenes were crossed to homozygous mutant flies (*se*^{Δ5} or *TM2/TM6B*, for transmission of *se* or *e* alleles, respectively) and transmission of base-edited alleles from the male parent determined by phenotypic assessment and Sanger sequencing. (B) Germline transmission of base-edited alleles as percentage of F2 flies exhibiting the respective phenotype (assessed from a total of *n* = 3 independent crosses). Transmission rates for each gRNA strain are reflective of the editing rates observed with Amp-Seq. (C and D) Alleles observed in F2 offspring with the e^{SPLICE} and e^{STOP_b} gRNAs, respectively. Shown are the protospacer (black) and PAM (red) of the target sites, as well as the editing window in grey (position 3-9 for evoAPOBEC1 and position -1-11 for evoCDA1), and target C residues with black triangles. Types and frequency of the observed alleles are shown. Only reference or alleles carrying C-to-T edits are observed. The position carrying a single-nucleotide polymorphism (SNP) is shown in orange and highlighted with an asterisk (where M is A or C).

To confirm that phenotypes were caused by the intended gene edits, we next performed Sanger sequencing of the e^{STOP_b} and e^{SPLICE} target sites in phenotypic offspring to assess the identity of the transmitted edits. In every instance (*N* = 42), we detected at least one base edit at the intended residue in one of the two target sites and did not observe any indels (Fig. 3, C and D), confirming that our assay accurately reflected the level of base editing in

the germ line. In line with the broader editing window of dCBE^{evoCDA1}, we more frequently observed alleles with bystander edits with this editor. For both dCBEs, we often recovered different base-edited alleles, some with an edit exclusively at the target C and others with additional edits at other residues in the target window (Fig. 3D). This demonstrates that screening of multiple offspring can be used to recover different alleles in the same cross. Overall,

pAct-dCBE^{evoAPOBEC1} more frequently led to precisely edited alleles without nonsilent bystanders, reflecting its narrower editing window, while *pAct-dCBE^{evoCDA1}* mediated a higher fraction of mutant offspring. Together, these results demonstrate efficient base editing in the *Drosophila* germ line to create heritable alleles.

Tissue-specific base editing

A key advantage of using model systems such as *Drosophila* is the possibility to study gene function in cells in their normal organismal context. To take full advantage of this possibility it is often necessary to restrict gene perturbations in space and time. This is particularly important for the study of essential genes, where lethality would otherwise preclude the study of gene function in later stages of life. We therefore aimed to develop a conditional base editing system. In *Drosophila*, conditional gene expression is most commonly achieved through the binary Gal4/upstream activation sequences (Gal4/UAS) system (41), in which a transgene under control of UAS is induced by a Gal4 "driver" (Fig. 5A). Thousands of Gal4 lines with tissue-specific expression patterns are publicly available.

To enable regulation of base editor expression by Gal4, we generated animals transgenic for *pUAS-dCBE^{evoAPOBEC1}* and *pUAS-dCBE^{evoCDA1}*, respectively, which were fertile and phenotypically normal. However, when crossing *pUAS-dCBE^{evoAPOBEC1}* animals to *hedgehog-Gal4* (*hh-Gal4*), *actin-Gal4* (*act-Gal4*), or *POU domain protein 2-Gal4* (*pdm2-Gal4*) driver lines to induce base editor expression, we did not obtain any viable offspring (Fig. 5A and fig. S9). In contrast, crosses with *pUAS-dCBE^{evoCDA1}* produced viable offspring with all three Gal4-drivers, although *pUAS-dCBE^{evoCDA1}* animals also carrying *pdm2-Gal4* or *hh-Gal4* frequently exhibited wing malformations indicative of reduced cell proliferation or excessive cell death (fig. S9). These findings suggest that high base editor overexpression, a known feature of the Gal4/UAS system, is toxic in *Drosophila* and that the *evoCDA1* deaminase domain is better tolerated than the *evoAPOBEC1* domain. The notion that toxicity is a result of excessive base editor expression is further supported by the observations that animals expressing either construct from the *act5C* promoter, which is known to mediate much lower levels of expression than the Gal4/UAS system (8), do not exhibit abnormal phenotypes and that wing phenotypes of *pdm2-Gal4 pUAS-dCBE^{evoCDA1}* animals are further enhanced by an additional copy of the base editor (fig. S9).

To more directly monitor adverse effects of base editor overexpression, we induced transgenes in the pouch of the wing imaginal disc with *pdm2-Gal4* and stained for cleaved *Drosophila* Caspase-1, which accumulates in apoptotic cells. Consistent with their organismal phenotype, overexpression of *pUAS-dCBE^{evoAPOBEC1}* leads to a strong, cell autonomous induction of apoptosis in all pouch cells, leading to a severe reduction in pouch size, while expression of *pUAS-dCBE^{evoCDA1}* with the same driver resulted in substantial but less widespread induction of cell death (Fig. 5, B to F).

The results described above highlight that CBE expression levels need to be carefully controlled. We therefore generated a modified UAS expression construct, using the *dCBE^{evoCDA1}* base editor, as it is better tolerated and more active than *dCBE^{evoAPOBEC1}*. First, we inserted a green fluorescent protein (GFP) coding sequence with a SV40 3' untranslated region (3'UTR) and flanked by flippase recognition target (FRT) sites between the UAS promoter and the *dCBE^{evoCDA1}* coding sequence (Fig. 5A). This *FRT-GFP* cassette

prevents expression of the base editor, allowing long-term maintenance of transgenic stocks that also carry a Gal4 driver without the risk of potentially harmful base editor activity. *dCBE^{evoCDA1}* expression can be induced by flippase (FLP) recombinase, which removes the GFP cassette, leaving behind a single FRT site. A wide range of transgenes encoding FLP under the control of tissue-specific or heat-shock (hs) inducible promoters are publicly available. We routinely generate gRNA lines in a genetic background that includes a *hs-FLP* transgene. Second, we cloned a medium sized upstream open reading frame (uORF, *u^M*) directly in front of the *dCBE^{evoCDA1}* coding sequence. We and others have shown that uORFs are a versatile method to reduce the levels of transgene expression (8, 42). To test whether the modified construct was better tolerated, we removed the *FRT-GFP* cassette in the germline through FLP expression and crossed the resulting *UAS-u^M-dCBE^{evoCDA1}* offspring to a *pdm2-Gal4* driver line. Offspring had normal levels of apoptosis in the wing pouch (Fig. 5, C and F) and developed wings of normal size and morphology (fig. S9), demonstrating that the improved construct design effectively prevented toxicity.

To test for conditional base editing with our improved construct, we generated stable fly lines carrying *UAS-FRT-GFP-FRT-u^M-dCBE^{evoCDA1}* and a Gal4 driver and crossed them to animals transgenic for gRNAs and a *hs-FLP* transgene. We then used a heat shock of limited duration to induce FLP expression in some, but not all, cells. This creates a genetic mosaic, in which cells with and without base editor expression can be identified by the absence or presence of GFP and compared side by side. We then stained the tissue for the protein encoded by the target gene. In wing discs of *pdm2-Gal4 UAS-FRT-GFP-FRT-u^M-dCBE^{evoCDA1} gRNA^{ctSTOP}* larva, Ct protein was efficiently lost from cells that lacked GFP expression, indicating highly efficient base editing (Fig. 5, G, H, and M). Similar results were obtained when targeting *evi* with the *gRNA^{eviSTOP}* transgene (Fig. 5, I, J, and M) or when using a different Gal4 driver and targeting the *smo* gene (Fig. 5, K, L, and M). In all cases, loss of protein was not observed in cells that retained GFP expression or did not express Gal4, indicating no leaky base editor expression in these cells.

Together, our results demonstrate that base editor activity can be effectively controlled in time and space by linking base editor expression to the binary Gal4/UAS and FLP/FRT systems and highlight the need to carefully optimize *dCBE* expression levels to avoid toxicity associated with excessive levels of base editor protein.

Efficient and precise creation of missense mutations

The predictable outcomes of base editing have made it a popular technique for inducing precise missense mutations, creating proteins with defined amino acid changes. However, in this application, nonsilent bystander edits cannot be tolerated. As a result, there are increased demands on target selection and gRNA design (see Discussion). To explore the feasibility of generating missense variants with base editing in *Drosophila*, we first elected to recreate two classical point mutations. *Notch^{Ax-E2}* is a gain-of-function allele of *Notch* (N) in which Histidine 1167 is substituted with Tyrosine (43, 44). *N^{Ax-E2}* animals are viable with morphological abnormalities, including a shortening of the L4 and L5 veins of the wing (45). We generated a transgenic fly line expressing a gRNA targeting the N locus around codon 1167 (Fig. 6A). The target C that upon base editing to T leads to the H1167Y substitution is located at position 2

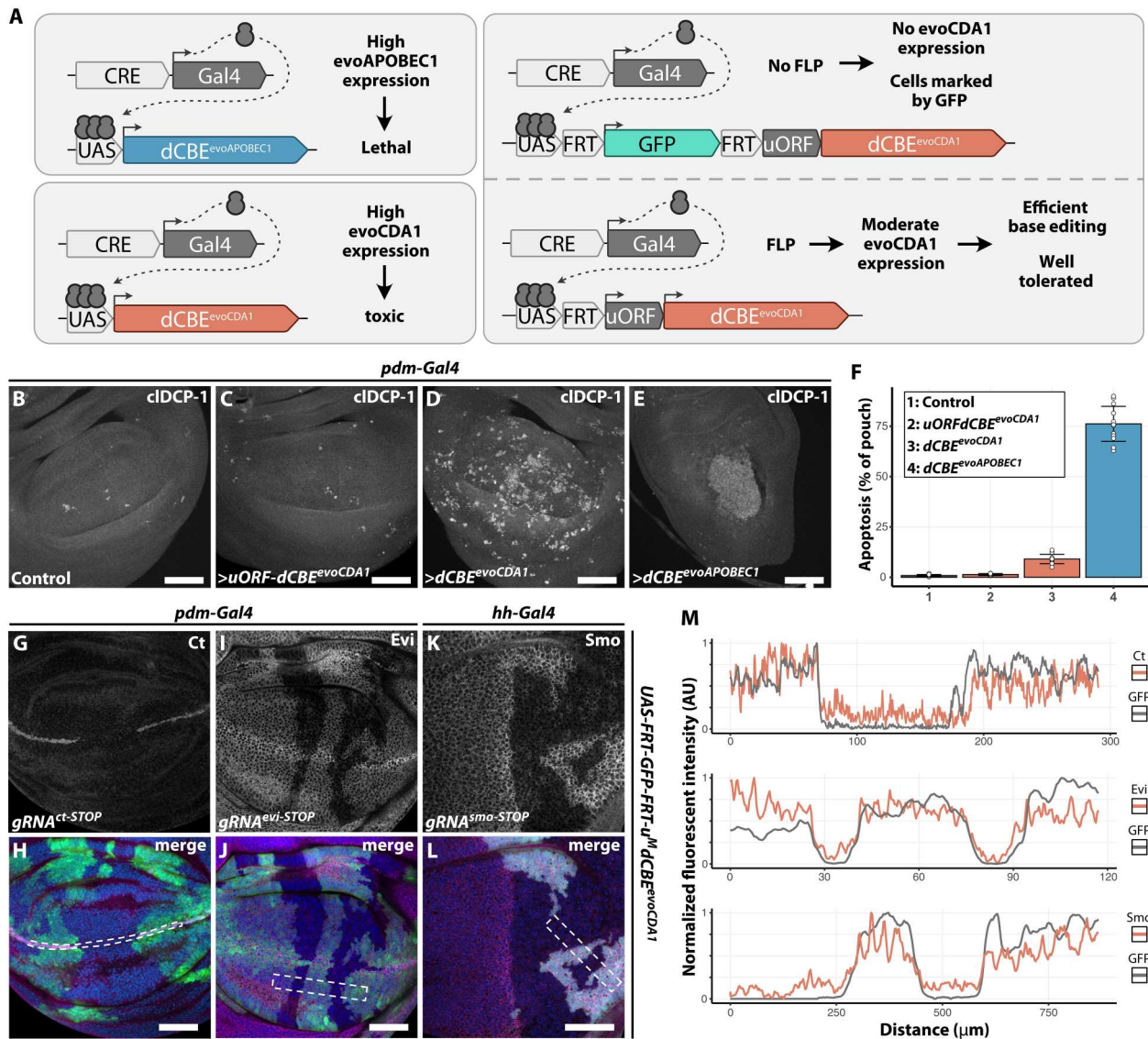


Fig. 5. Efficient tissue-specific base editing with evoCDA1. (A) Schematic of the experimental strategy. Left: Expression of either editor directly downstream of Gal4/UAS results in high expression levels and lethality or toxicity (CRE = cis regulatory element). Right: A FRT-GFP cassette makes dCBE expression conditional on FLP recombination. Cassette excision creates base editor expressing cells marked by the absence of GFP. In addition, expression is attenuated by a uORF. (B to E) High base editor overexpression is toxic. Maximum intensity projections of 16 confocal images spanning 5 μm of the basal surface of the wing imaginal disc, stained for cIDCP-1, which accumulates in apoptotic cells. *pdm2-Gal4* is expressed in the pouch (central area) of the wing disc. Only few apoptotic cells are observed in control discs (B, Gal4 only) and in samples expressing UAS-*uORF dCBE^{evoCDA1}* (C, GFP cassette removed before experiment). Substantially more cells undergo apoptosis in *pdm2-Gal4* UAS-*dCBE^{evoCDA1}* discs (D). Most cells of the wing pouch are cIDCP-1 positive in *pdm2-Gal4* UAS-*dCBE^{evoAPOBEC1}* wing discs, causing reduced tissue size (E). (F) Quantification of apoptotic area (cIDCP-1 staining) in the pouch of the indicated genotypes (mean ± SD, points representing individual discs). (G to L) Efficient tissue-specific base editing in marked cell populations. Immunofluorescent staining of wing discs of indicated genotype. The GFP cassette was removed through limited expression of FLP recombinase 48 hours before sample preparation. In (G) and (J), expression is driven by *pdm2-Gal4* in the pouch of the wing disc and in (K) and (L) with *hh-Gal4*, which is active in the posterior half of the tissue. (merge images GFP green, DNA blue, endogenous protein magenta). (M) Line intensity profiles of GFP and antibody staining of the indicated proteins. To help comparison, intensities were normalized and plotted along the long axis of the dashed boxes shown in (H), (J), and (L). Scale bars, 50 μm. AU, arbitrary units.

within the protospacer, with the closest cytidine that could lead to an undesired missense mutation located at position 18 (Fig. 6A). All animals ($N = 69$) transgenic for *pAct-dCBE^{evoCDA1}* and the *N^{Δx-E2}* gRNA had wings with shortened L4 and L5 veins, indicating highly efficient base editing inducing the gain-of-function allele (Fig. 6B). In contrast, animals expressing *pAct-dCBE^{evoAPOBEC1}* and the *N^{Δx-E2}* gRNA were morphologically normal. Consistent

with these phenotypic observations, amplicon sequencing of the target locus revealed $95.5 \pm 0.6\%$ C-to-T editing at the target residue with *dCBE^{evoCDA1}*, while only $0.2 \pm 0.1\%$ editing was detectable with *dCBE^{evoAPOBEC1}* (fig. S10). These results are consistent with the different editing windows of each base editor (Fig. 2C and fig. S10). We only detected $0.2 \pm 0.3\%$ C-to-T editing at the closest potential nonsilent bystander edit with *dCBE^{evoCDA1}* (fig.

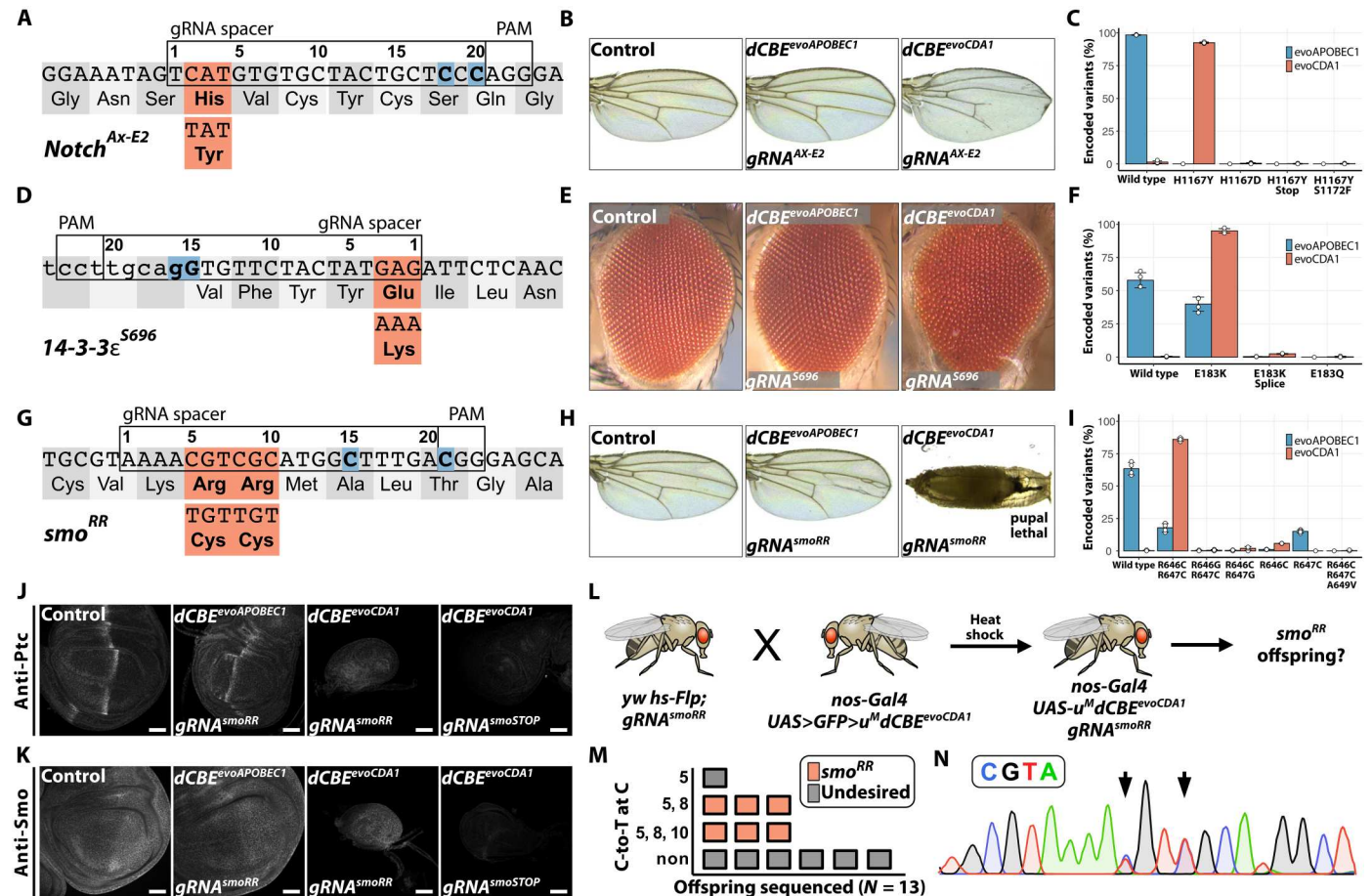


Fig. 6. Efficient induction of missense mutations with dCBEs. (A) Schematic of the *N^{AX-E2}* gRNA target site. The desired mutation is highlighted in red, nonsilent bystanders in blue. Nonhighlighted C residues result in silent edits. (B) Wing phenotypes of the indicated genotypes. Wings of *pAct-dCBE^{evoCDA1}gRNA^{AX-E2}* animals have shortened L4 and L5 veins. (C) Quantification of reads encoding the indicated protein variant. To exclude sequencing errors, reads comprising at least 0.5% of total reads are considered (mean $\pm$ SD, points representing individual replicates). (D) Schematic as in (B) for the *14-3-3* locus. (E) Eye phenotypes induced by *gRNA¹⁴⁻³⁻³*. (F) Quantification of sequencing reads encoding protein variants as in (C). (G) Schematic as in (A) for the *smo* locus. (H) Phenotypes induced by editing with *gRNA^{smoRR}*. (I) Quantification of sequencing reads encoding protein variants as in (C). (J) Staining of wing imaginal discs of the indicated genotype for Ptc protein, expressed in a stripe along the anterior-posterior boundary in wild type. Discs expressing *dCBE^{evoCDA1}* and either *gRNA^{smoRR}* or *gRNA^{smoSTOP}* fail to express *ptc*. (K) Staining of wings discs for Smo protein. Base editing with *gRNA^{smoRR}* does not impair expression of Smo. Scale bars, 50 μ m. (L) Schematic for tissue-specific germline editing of *smo*. *nos-Gal4 UAS-FRT-GFP-FRT-u^MdCBE^{evoCDA1} hs-Flp gRNA^{smoRR}* animals were heat shocked at first and third instar larval stages for 40 min at 37°C. Adult males were crossed to wild-type females, and offspring were analyzed by sequencing. (M) Summary of Sanger sequencing results. Of 13 flies analyzed, 6 had base edits exclusively in the codons encoding the two arginine residues (*smoRR*), while 7 had either no mutations or only received mutations at one of the arginine codons (classified as "undesired"). (N) Example Sanger sequencing trace. Highlighted are the two positions at which the paternal base-edited allele and the maternal wild-type allele overlap.

S10). Accordingly, analysis of individual sequencing reads (only including alleles covered by at least 0.5% of total reads, see Materials and Methods) revealed that base editing with *dCBE^{evoCDA1}* resulted in $92.5 \pm 0.6\%$ reads encoding the precise H1167Y allele, while only $2.2 \pm 1.9\%$ encoded other alleles, including wild type or those encoding undesired mutations (Fig. 6C).

Next, we performed base editing to recreate the *14-3-3^{S-696}* allele, a dominant-negative missense mutation that substitutes glutamic acid 183 with lysine in *14-3-3 ϵ* (Fig. 6D) (46, 47). *14-3-3^{S-696}* flies are subviable and have defective eye morphology, as well as defects in oocyte determination and polarization (47, 48). Flies transgenic for *pAct-dCBE^{evoCDA1}* and a *14-3-3^{S-696}* gRNA were subviable and had irregular eye morphology, while *pAct-*

dCBE^{evoAPOBEC1}/gRNA^{14-3-3 ϵ S-696} animals were viable and had only rare irregular eye patches (Fig. 6, D and E). Amp-Seq of *dCBE^{evoCDA1}* animals revealed that $95.1 \pm 1.6\%$ of reads encoded the precise E183L allele, with the remainder being either wild type ($0.4 \pm 0.3\%$) or containing undesired mutations ($2.7 \pm 0.8\%$) (Fig. 6F). In contrast, only $40 \pm 5.3\%$ of reads encoded precisely the intended mutation when editing was performed with *dCBE^{evoAPOBEC1}*.

Having established efficient and precise missense mutagenesis with base editing, we set out to create a mutation that has so far not been created in *Drosophila*. Recently, it was shown in mouse cells and zebrafish that the atypical G protein-coupled receptor Smo controls Hedgehog (Hh) signaling via a protein kinase A

pseudosubstrate site in its cytoplasmic tail (49). While the central di-arginine motif of this site is also present in *Drosophila* Smo, its importance for Hh signaling has so far not been investigated. We therefore created a gRNA transgene targeting the *smo* locus at the respective position (referred to as gRNA^{smoRR}; Fig. 6G) and combined it with each dCBE. *pAct-dCBE^{evoCDA1}/gRNA^{smoRR}* animals died at pupal stage, a phenotype consistent with strong inhibition of Hh signaling, while *pAct-dCBE^{evoAPOBEC1}/gRNA^{smoRR}* flies were viable with occasional wing phenotypes, suggesting disruption of Hh signaling only with low penetrance (Fig. 6H). Consistent with these phenotypes, we found that $88.3 \pm 3.2\%$ of reads encoded *smo* alleles that precisely substituted the two arginines without non-silent bystander mutations when editing was performed with dCBE^{evoCDA1}, while only $18.3 \pm 4.1\%$ of reads had the desired combination of mutations with dCBE^{evoAPOBEC1}. Furthermore, wing imaginal discs expressing dCBE^{evoCDA1} and gRNA^{smoRR} failed to grow and did not express the Hh target gene patched (*ptc*), while *pAct-dCBE^{evoAPOBEC1}/gRNA^{smoRR}* discs had normal size and only slightly abnormal *ptc* expression (Fig. 6J). Discs of both genotypes still expressed normal levels of Smo protein, indicating that the phenotype is caused by a precise manipulation of the di-arginine motif and not nonsense mutations interfering with Smo expression or stability (Fig. 6K).

Last, we wanted to test whether the high efficiency of dCBE^{evoCDA1} would also allow the creation of precise, heritable missense alleles with minimal effort. However, germline transmission of *smo^{RR}* mutations from *pAct-dCBE^{evoCDA1}/gRNA^{smoRR}* animals is not possible, as these animals die before eclosion. We therefore tested whether we could use our optimized tissue-specific dCBE^{evoCDA1} expression construct to perform germline transmission of loss-of-function alleles in this essential gene. We combined *UAS-FRT-GFP-FRT-*u^M*dCBE^{evoCDA1}* with *nanos-Gal4VP16*, a driver specifically expressed in the *Drosophila* germline, and crossed these to *hs-FLP; gRNA^{smoRR}* flies (Fig. 6L). After heat shock (see Materials and Methods), the resulting offspring was viable and phenotypically normal, indicating successful exclusion of base editing from somatic cells. We then crossed these animals and sequenced the *smo* locus of 13 randomly selected offspring. We found that six of these inherited *smo* alleles encoding mutations exclusively in the two arginine codons (Fig. 6, M and N).

In summary, these results demonstrate how the remarkable efficiency and precision of our base editing tools and in particular dCBE^{evoCDA1} can be harnessed to induce precise mutations in endogenous genes to produce protein variants in somatic or germline cells.

DISCUSSION

In this study, we establish efficient CBEs in *Drosophila* for gene inactivation and the generation of missense mutations through C-to-T editing. The system is composed of genetically encoded CBEs and gRNAs to create premature stop codons and disrupt splice sites or create alleles encoding precise amino acid substitutions. A base editor containing the evoCDA1 deaminase domain, which has high activity at temperatures below 30°C, routinely mediates the generation of biallelic nonsense or missense alleles in most cells of the animal. In addition, the exceptionally high product purity of base editing in *Drosophila* and its compatibility with existing

strategies for gRNA multiplexing further contribute to the very high efficiency of this system.

Efficiency of gene disruption experiments with CRISPR nucleases in bulk cell populations and entire organisms is usually curbed by limited gRNA activity and the generation of function-retaining mutations at the target site (4, 7). Multiplex base editing presents a broadly applicable alternative to precisely install loss-of-function alleles. The generation of stop codons is hereby more widely applicable, as typically there are many more positions in a gene where it is possible to convert codons to stop codons than there are targetable splice sites, in particular in organisms with compact genomes (14, 15). For example, we were not able to design SPLICE gRNAs against the small *se* gene. Moreover, when targeting splice sites, splice acceptors of small exons should be avoided given the tendency to result in exon skipping, as previously described (16, 17) and also evidenced by the *f^{SPLICE_a}* gRNAs in our study. In contrast, premature stop codons are expected to have severe functional consequences on gene function, in particular when placed in the 5' portion of the gene, where they are also more likely to cause nonsense mediated mRNA decay (50, 51). However, the possibility of stop codon read-through and functionality of polypeptides encoded 5' of the first base-edited site should be considered when designing and analyzing such experiments. For both strategies, computational tools are available for the identification of suitable gRNAs (14, 17) (see also Materials and Methods). Notably, such gRNAs can in many cases also be used in combination with existing Cas9 nucleases to create mutations by error-prone double-strand break repair, should both nonhomologous end joining (NHEJ)-mediated indels and base editing-mediated C-to-T mutations be of interest.

Another powerful application of base editing is the induction of precise missense mutations to analyze the functional consequences of amino acid substitutions on protein function. We demonstrate here that the remarkable efficiency of the evoCDA1 domain allows the near homogeneous generation of such mutations in entire tissues or organisms, readily revealing their mutant phenotype. There are no other technologies available today that would allow the same with similar efficiency. While classical homology-directed repair of double-strand breaks induced by CRISPR nucleases suffers from the very high co-occurrence of undesired editing outcomes, the efficiency of prime editing in somatic cells of the fly is so far substantially lower than what we observe here with dCBE^{evoCDA1} (52). However, the number of missense alleles that can currently be created with base editing is limited due to the defined nature of mutations (e.g., only C-to-T with CBEs), the required position of the editing site relative to a protospacer adjacent motif (PAM), and the presence of potential nonsilent bystander edits in the editing window. In this respect, the wide editing window of evoCDA1 is an ambivalent feature of this base editor, as it increases the genomic target space, as more C residues can be accessed from any given PAM, but at the same time substantially reduces the number of mutations that can be generated without nonsilent bystander edits, reducing its utility for generating clean missense mutations. In the future, we anticipate a substantial expansion in the number of targetable sites through the generation of additional base editors, such as adenine base editors, for *Drosophila*. Furthermore, we expect that combining the highly efficient evoCDA1 domain with Cas9 variants with relaxed PAM requirements will further contribute to this increase (53).

Our work reveals the crucial role of temperature in determining base editing efficiency with different CBEs. While the evoAPOBEC1 deaminase domain shows rapidly declining activity at temperatures below 30°C, evoCDA1 is better adapted to this temperature range, likely reflecting the fact that these deaminase domains originate from an endothermic and ectothermic organism, respectively. While the fact that different enzymes are optimized for different temperatures is generally recognized, temperature is rarely a tested parameter during the development of gene editing tools. As a result, the optimal temperature range for most genome editors is not known. Prior work has revealed that variants of the Cas12a nuclease derived from different bacterial species vary markedly in their activity at lower temperature (54, 55). Our study expands this observation to deaminases used for precision genome engineering. Outside of applications in mammals, many of the most important uses of CRISPR genome engineering take place in organisms that do not tolerate longer term exposure to temperatures of 37°C. For example, important disease vectors, such as mosquitos of the genus *Aedes* and *Anopheles* that transmit malaria, dengue, and other diseases, are optimally adapted to temperatures between 20° and 30°C, with an upper limit of 34°C (56, 57). Likewise, the optimal growth temperatures for the majority of crop plants are below 30°C (58). It is therefore important to develop genome engineering tools optimized to different climates to realize the full potential of CRISPR technology.

One unexpected finding of our study is the fact that base editing in *Drosophila* leads to much fewer undesired by-products in the form of indels or C-to-G or C-to-A edits at the gRNA target site compared to many other species. This is likely due to system specific differences in DNA repair. Indel by-products during cytosine base editing mostly arise from error-prone repair of abasic sites that are generated by the UNG enzyme (10, 24). To limit such events, CBEs are typically linked to one or more UGI domains (24, 25). *Drosophila* and other holometabolous insects are naturally missing a *UNG* gene (33) and thus mimic human *UNG* knockout cells, which had previously been shown to essentially eliminate indels and C-to-G/A editing (24). The fact that we did observe some indel formation, in particular with the highly active evoCDA1 domain, suggests that additional DNA repair enzymes recognize intermediate base editing products and contribute to the formation of undesired editing outcomes. Indeed, fusion of various glycosylases and other BER components to base editors has previously been shown to lead to substantial diversification of editing outcomes (39), supporting the notion that *UNG* is the major but not sole source of CBE by-products.

The data presented here also highlight organismal toxicity as a potential limitation of current base editing technology. While base editing does not induce genotoxic DNA double-strand breaks, it does rely on the expression of constitutively active deaminase domains, which on their own act indiscriminately on single-stranded DNA and in some cases RNA (59, 60). Specificity of such systems relies on the ability to concentrate an active base editor at the target site through RNA-guided DNA binding of Cas9 while limiting base editor concentration in the rest of the cell. Accordingly, a previous study has shown that limiting CBE expression to the level of housekeeping genes abrogates the Cas9-independent DNA and RNA off-targeting observed with transfection-mediated overexpression (61). Furthermore, a recent study in mice demonstrated the induction of off-target mutations in DNA and RNA with APOBEC-based base

editors that correlated with CBE expression levels (62). We expand these findings to another model system and show that CBEs expressed at moderate levels from a housekeeping promoter (*act5c*) or attenuated via an uORF are well tolerated in *Drosophila*, while high overexpression with the binary expression system Gal4/UAS gives rise to severe side effects. The level of toxicity is strongly dependent on the deaminase domain. While *UAS-dCBE^{evoAPOBEC1}* resulted in lethality when expressed under the control of an *act-*, *pdm2-*, or *hh-Gal4* driver, *UAS-dCBE^{evoCDA1}* was better tolerated under the same conditions, despite the higher activity of this editor. It is tempting to speculate that this at least in part is due to the fact that APOBEC1, but not CDA1, exhibits RNA off-targeting in human cells (59, 63). Whether this is also true in *Drosophila* and whether the two base editors differ in their DNA off-target profile in flies will be an important question to address in the future. Going forward, the use of engineered deaminase domains with reduced gRNA-independent off-targeting (63, 64) might further reduce the toxicity of base editors in *Drosophila* and other species. Likewise, recently described TadaA-based CBEs have been shown to minimize the occurrence of gRNA-independent DNA and RNA off-targeting (65, 66). Last, future work will focus on the characterization of gRNA-dependent off-targets and to what extent they might potentially contribute to the organismal tolerability or cause false-positive phenotypes.

In summary, our study describes highly efficient base editing tools for the installation of loss- and gain-of-function alleles in somatic cells or the germline of *Drosophila*. These systems will be useful for the efficient establishment of mutant fly lines or testing of gene function in the F1 generation. Base editor-mediated gene inactivation will be particularly advantageous in scenarios where function-retaining indels are problematic, knowledge of the type of edit is desired, deleterious consequences of DSBs are particularly problematic, or when polymerase chain reaction (PCR) bias would obscure sequencing-based analysis of target sites of different length. Our work also highlights several avenues to further improve CRISPR technology, such as optimization of temperature range and expression levels and leveraging the natural diversity in DNA repair between different systems. Thus, the tools and strategies described herein should have broad usefulness for genetic experiments in *Drosophila* and other multicellular organisms and inform future developments of CRISPR technology.

MATERIALS AND METHODS

Plasmid construction

A list of oligonucleotides and gBlock dsDNA fragments used in this study is provided in table S1. PCRs were performed using Q5 Hot-start 2x master mix (New England Biolabs). Restriction digests were conducted for 2 hours at 37°C. The linearized plasmid backbones and PCR products were analyzed on agarose gels and purified using the QIAquick gel extraction kit (Qiagen). Assembly of DNA fragments was either performed with isothermal assembly or Goldengate-like reactions (see details on individual plasmids below). Reactions were transformed into Stellar competent cells (Takara) and plated on lysogeny broth (LB) agar plates supplemented with carbenicillin (100 µg/ml). Individual colonies were grown in LB broth supplemented with carbenicillin (100 µg/ml) for 16 hours at 37°C and 180 rpm. The plasmid DNA was extracted using the

QIAprep spin miniprep kit (Qiagen), and newly inserted sequences were verified by Sanger sequencing.

pAct-dCBE^{evoAPOBEC1}

DNAs encoding the N- and C-terminal CBE components were ordered as gBlocks (Integrated DNA Technologies) and PCR-amplified. The Cas9(D10A) coding sequence was generated through PCR amplification from plasmid *pAct-Cas9* (34) using primers *Cas9D10Afw/Cas9D10Arev*. The purified PCR products were assembled into Eco RI-HF/Xho I-digested *pAct-Cas9* (34) by In-Fusion cloning (Takara).

pAct-dCBE^{evoCDA1}

A gBlock encoding the evoCDA1 domain and Cas9 N terminus until the Nde I restriction site in *pAct-dCBE^{evoAPOBEC1}* was amplified using primers *oRMD204/oRMD205*. A second PCR was conducted on *pAct-dCBE^{evoAPOBEC1}* with primers *oRMD158/oRMD206* to generate a fragment with overlaps to the upstream NdeI site and 5' end of the evoCDA1 domain. The PCR fragments were assembled into NdeI-digested *pAct-dCBE^{evoAPOBEC1}* using NEBuilder HiFi DNA assembly (New England Biolabs).

pUAS-dCBE^{evoAPOBEC1}

The dCBE^{evoAPOBEC1} coding sequence was amplified from the *pAct-dCBE^{evoAPOBEC1}* plasmid using primers *oRMD134/oRMD135*, and the purified product was assembled into EcoRI-HF/XbaI-digested *pUASTattB* backbone (67) using NEBuilder HiFi DNA assembly (New England Biolabs).

pUAS-dCBE^{evoCDA1}

The dCBE^{evoCDA1} coding sequence was amplified from the *pAct-dCBE^{evoCDA1}* plasmid using primers *oRMD134/oRMD135*, and the purified product was assembled into EcoRI-HF/XbaI-digested *pUASTattB* backbone (67) using NEBuilder HiFi DNA assembly (New England Biolabs).

pUAS-FRT-GFP-FRT-uMdcBE^{evoCDA1}

A cassette encoding a GFP and SV40 3'UTR flanked by FRT sites and an uORF encoding the first 38 amino acids of β -galactosidase was cloned into *pUAS-dCBE^{evoCDA1}* in a two-step cloning. First, a KpnI site was introduced in front of the *dCBE* coding sequence with extension PCR of two amplicons generated with primers *insertKpnIfwd1/rev1* and *insertKpnIfwd2/rev2* and assembled with an *UAS-dCBE^{evoCDA1}* backbone created by digestion with NheI and NdeI. Afterward, the plasmid was linearized by KpnI digestion, the *GFP-uORF* cassette was PCR-amplified from plasmid *pUAS-FRT-GFP-FRT-uMCas9* (8) with primers *evoCDA1GFPuORFfwd/rev*, and both fragments were assembled using the NEBuilder HiFi DNA assembly (New England Biolabs) to yield the final plasmid.

pCFD5-gRNAs

gRNA vectors (1 $\times$ and 2 $\times$) were constructed as previously described (8). For plasmids encoding single gRNAs, oligos encoding the protospacer sequence were annealed and ligated into the pCFD5 plasmid. For 2 $\times$ gRNA plasmids, forward and reverse primers encoding BbsI handles and the spacer sequences of gRNAs #1 and #2 were used in PCRs on plasmid *pCFD6* (35). The purified reaction products were pooled at equal molarity and assembled into *pCFD5* in a Goldengate reaction using BbsI-HF and T4 Ligase (both New England Biolabs).

CBE gRNA design

The iSTOP R package (14) and SpliceR webtool (17) (<https://moriaritylab.shinyapps.io/splicer/>) were used to identify potential

STOP and SPLICE gRNAs, respectively. Target sites for experimental interrogation were then selected on the basis of protospacer composition (40 to 60% GC content, balanced nucleotide representation), relative location in the coding sequence of the gene (first half), and position of the target residue within the editing window (5-7). In cases when no such two target sites were available, gRNAs at more 3' regions in the gene and target residues at more sideward position of the editing window (4-9) were chosen. Spacer sequences of all gRNAs used in this study are provided in table S2.

Drosophila strains and handling

General husbandry

All *Drosophila* strains generated and/or used in this study are listed in table S3. Flies were kept at 50 $\pm$ 10% humidity in a 12-hour light/12-hour dark cycle. The temperatures at which individual experiments were conducted are indicated in the figures. Temperature was regularly checked with thermometers.

Genomic DNA preparation

Materials from individual flies (generally heads, if not otherwise indicated) were collected in 20 μ l of squishing buffer [10 mM tris-HCl (pH 8), 1 mM EDTA, 25 mM NaCl, and proteinase K (200 μ g/ml)]. For genomic DNA (gDNA) extraction from larvae, a dissection was performed to remove gut and fat tissue. The tissue was then disrupted in a Bead Ruptor (Biovendis) for 20 s at 25 Hz. The samples were incubated for 30 min at 37°C for proteinase K digestion and heat-inactivated at 95°C for 3 min.

Transgenesis and genotyping

Transgenesis was conducted using the PhiC31/attP/attB system (67). Before injections, plasmid DNA was purified using the QIAquick PCR purification kit (Qiagen) and normalized to 250 ng/ μ l in nuclease-free water. For pooled injections, an equimolar ratio of all plasmids was pooled. Microinjections were performed into *y[1] M{vas-int.Dm}ZH-2A w[*]*; (*P{y[+7.7]CaryP}attP40*) blastoderm embryos for second chromosomal integration and on *y[1] M{vas-int.Dm}ZH-2A w[*]*; (*P{y+-attP-3B}VK00033*) embryos for third chromosomal integrations following standard procedures. The embryos were then raised at 18°C until reaching the larval stage (approximately 48 hours). The animals were then transferred to 24°C and raised to adulthood and individual flies crossed to *P{ry[+7.2] = hsFLP}1, y[1] w[1118]*; *Sp/CyO-GFP* or *w[*]*; *Sb/TM6* balancer flies for second and third chromosomal integrations, respectively. Transgenic offspring were identified by eye color and then crossed against appropriate balancer flies to generate a stable genetic stock.

If multiple gRNA plasmids were injected in a pool, transgenic adults were subjected to genotyping after having produced sufficient numbers of offspring. To this end, 2 μ l of gDNA was used in PCRs with primers *pCFD8genofwd5/pCFD8genorev6*, and the purified PCR product subjected to Sanger sequencing with primer *pCFD8sequonly*.

Germline transmission of base edits

Transmission of viable alleles with pAct-CBE

P{ry[+7.2] = hsFLP}1, y[1] w[1118]; *pAct-dCBE^{evoAPOBEC1} or evoCDA1/+*; *pU6:3-gRNAs/+* males obtained in the F1 generation were crossed to *w[*]*; *Bl/CyO*; *TM2/TM6B* virgins for gRNAs targeting *ebony* and *se Δ 5* virgins for gRNAs targeting *sepia*. The percentage of phenotypically *ebony* or *sepia* flies was then determined in the F2 generation. To exclude F2 animals that inherited both genome

editing transgenes from their male parent, we only assessed animals lacking at least one of the two transgenes (determined by fainter eye color). For transmission of *sepia* alleles, this exclusion was achieved on the basis of the presence of somatic *sepia* mosaics that were observed when an F2 animal inherited both genome editing components and a functional *sepia* allele in their germline. For molecular determination of the transmitted edits, gDNA was extracted from F2 animals, and the target loci were amplified by PCR. The purified reaction products were submitted to Sanger sequencing, and the transmitted edits were identified by visual inspection of the sequencing chromatograms.

Transmission of *smo^{RR}* alleles by germline-restricted base editing

Males of the genotype *w*[1118]; *UAS-FRT-GFP-FRT-u^MdCBE^{evoCDA1}/CyO*; *nanos-Gal4VP16 / TM6B* were crossed to *P{ry[+t7.2]} = hsFLP*1, *y*[1] *w*[1118]; *U6:3-gRNA-smo^{RR}* virgin females, and crosses were reared at 29°C. Offspring was subjected to two 40-min heat shocks at 37°C when approximately at first instar larval stage and mid pupal stage to induce *hs-Flp* expression. Male offspring of the genotype *P{ry[+t7.2]} = hsFLP*1, *y*[1] *w*[1118]; *U6:3-gRNA-smo^{RR}/UAS-FRT-GFP-FRT-u^MdCBE^{evoCDA1}*; *nanos-Gal4VP16/+* were then crossed to *w*[*] virgin females. Randomly selected offspring was tested for inheritance of *smo^{RR}* alleles by extracting gDNA, amplifying the *smo* locus by PCR and Sanger sequencing.

Sanger sequencing of base editing target sites

Genomic loci were amplified using primers located at least 50 nucleotides 5' and 3' of the target sites of a given gRNA strain. The reactions were purified with paramagnetic beads and submitted for Sanger sequencing. For quantification of editing rates from sequencing chromatograms, the Base Editing Analysis Tool was used (68).

Quantification of genome editing outcomes by deep Amp-Seq

Library preparation

Targeted deep sequencing of CRISPR edits was performed using a dual PCR-based strategy similar to previously described (53). Briefly, in a first PCR, genomic target sites were PCR-amplified (Q5 Hot Start High-Fidelity DNA Polymerase, New England Biolabs) using primers encoding Illumina adapter sequences (table S4). Reaction clean-up and size selection were performed using paramagnetic beads prepared as previously described (69). The concentrations of the purified reaction products were determined on a DropSense 96 (Trinean), and approximately equal amounts of different genomic amplicons were then pooled and subsequently diluted in nuclease-free water by a factor of 10. A second PCR was then performed on the diluted amplicon pools with primers adding Illumina indexes and P5/P7 sequences (table S4). The reactions were purified and size-selected using paramagnetic beads and then quantified on a DropSense 96 (Trinean). Equal amounts of all libraries were combined into a final multiplex, which was quantified with a Qubit 1.0 fluorometer (Thermo Fisher Scientific) and then sequenced on an Illumina MiSeq (150-bp paired-end reads) by the High-Throughput Sequencing Unit of the Genomics and Proteomics Core Facility (German Cancer Research Center, DKFZ).

Data analysis

For data analysis, CRISPResso2 (70) was run in pooled mode with the following parameters: *-w 20 --cleavage_offset -10 --base_editor_output --min_reads_to_use_region 100*. The sequences of the genomic amplicons are provided in table S5. Nucleotide conversion rates at individual nucleotides were extracted from the Quantification_window_nucleotide_percentage_table.txt files. The number of reads containing insertions or deletions was extracted from the CRISPResso_quantification_of_editing_frequency.txt files, and the percentage of reads containing indels was then calculated as $[\# \text{ reads containing indels}] / [\# \text{ reads aligned}]$. For amplicons containing single-nucleotide polymorphisms (SNPs) in their quantification window, CRISPResso2 was run with both alleles to avoid false classification of SNPs as NHEJ events. Indel sizes were extracted from the Indel_histogram.txt files. For quantification of alleles encoding precise missense mutations, allele frequency was extracted from the Allele_frequency_table_around_sgRNA.txt tables. To avoid missannotation of alleles representing sequencing errors, only alleles detected by at least 0.5% of all reads were considered.

CBE efficiency prediction with BE-HIVE

BE-Hive (36) (<https://crisprbehive.design/>) was run in single mode and provided with 20 nucleotide sequence contexts upstream and downstream of the spacers (i.e., a total of 60 nucleotides of the genomic target region). "evoAPOBEC1, mES" was selected as base editor and cell type setting. The Z scores predicted by the model were correlated with the observed base editing rates.

Immunohistochemistry

Third instar wandering stage larvae were dissected in ice-cold phosphate-buffered saline (PBS) using standard procedures. Larvae were fixed in 4% paraformaldehyde for 25 min at room temperature. Larvae were washed three times 15 min with PBT (1× PBS with 0.3% Triton X-100) and blocked for 30 min in PBT containing 1% normal goat serum. Samples were incubated with first antibodies {rabbit anti-Evi [1:500; (71)], mouse anti-smo [1:50; Developmental Studies Hybridoma Bank (DSHB)], mouse anti-ct [1:20; DSHB], and rabbit cleaved DCP-1 [1:500]} overnight at 4°C. Following three 15-min washes with PBT, samples were incubated with secondary antibodies (Invitrogen, goat anti-rabbit or goat anti-mouse conjugated to Alexa fluorophores, 1:400 in PBT containing 4',6-diamidino-2-phenylindole) for 2 hours at room temperature. Larvae were washed three times for 15 min at room temperature, and wing imaginal discs were mounted in Vetrashield (Biozol).

Image acquisition and processing

Microscopy images were acquired on a Leica LSM SP8 confocal microscope with a 40× numerical aperture 1.3 oil immersion lens in sequential scanning mode. Samples of the same experiment were imaged in the same session. Images were processed and analyzed using Fiji.

Supplementary Materials

This PDF file includes:

Figs. S1 to S11

Tables S1 to S5

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